# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Built and deployed **machine learning models** (classification, regression, clustering) using Python (scikit-learn, pandas, NumPy), generating predictive insights from large structured datasets.
- Designed **feature engineering and data preprocessing pipelines**, improving model accuracy by 18% through structured experimentation and cross-validation techniques.
- Developed a **customer churn prediction model** using gradient boosting and logistic regression, achieving 87% AUC and enabling proactive targeting of at-risk customer segments.
- Built and orchestrated **model training and evaluation pipelines** with MLflow experiment tracking, ensuring reproducibility and version control across model iterations.
- Integrated **model outputs into backend services** via REST APIs built in Java Spring Boot, enabling real-time inference and decision-support for operational teams.
- Worked with **PostgreSQL and MongoDB** to design efficient data extraction workflows and optimise pipeline throughput by 40% for large-scale data science projects.
- Collaborated with data engineers, analysts, and business stakeholders in **Agile/Scrum** sprints to translate business problems into machine learning solutions.
- Communicated model performance and business impact clearly to **technical and non-technical audiences**, driving informed adoption of data science recommendations.

## SKILLS

**Machine Learning** – scikit-learn, regression, classification, clustering, gradient boosting, model evaluation.

**Programming Languages** – Python (advanced), SQL, Java, R.

**Python Ecosystem** – pandas, NumPy, matplotlib, seaborn, Flask, FastAPI.

**Databases** – PostgreSQL, MySQL, MongoDB; data extraction and pipeline design.

**Cloud & MLOps** – AWS (S3, EC2, SageMaker basics), Docker, MLflow, Jenkins.

**Development Practices** – Agile/Scrum, Git, experiment tracking, model versioning.

**Tools** – Jupyter Notebook, VS Code, Tableau, Power BI.

## PROJECTS

### Customer Churn Prediction Model (HCL Technologies)

- Built a gradient boosting classification model to predict customer churn risk from structured transactional data, achieving 87% AUC and enabling proactive customer retention targeting.
- Designed feature engineering pipeline including categorical encoding, normalisation, and lag-based temporal features, improving model accuracy by 18% over baseline.
- Deployed model outputs via a FastAPI REST endpoint consumed by the CRM team for real-time risk scoring at point-of-contact.

### Sales Forecasting & Demand Analysis Pipeline (HCL Technologies)

- Built an end-to-end forecasting pipeline using Python and scikit-learn to predict monthly demand across product categories, reducing inventory overstock by 12%.
- Automated data extraction from PostgreSQL and model retraining with Jenkins, ensuring weekly updated forecasts without manual intervention.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)